GEBZE TECHNICAL UNIVERSITY

Computer Engineering Department

CSE 396 Module Documentation

Çılgın Kurye

Module M3: Physics and Collision

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1. Purpose

M3 moves the motorcycles. It turns throttle, brake and steering into speed, direction and position. It finds collisions, decides when a motorcycle falls, and adds the effect of obstacles and earthquakes to the motion. The same code runs in two places: on the server, where its result is the real one, and on the phone, where it guesses the player's own motorcycle before the server answers.

Runs on	Main computer inside the game server; phone browser for prediction only
Technology	TypeScript library; our own simple 2D arcade physics, no ready physics engine; circle and box collisions on a uniform grid; seeded random numbers
Requirements	R3 R4 R6 R7 R11

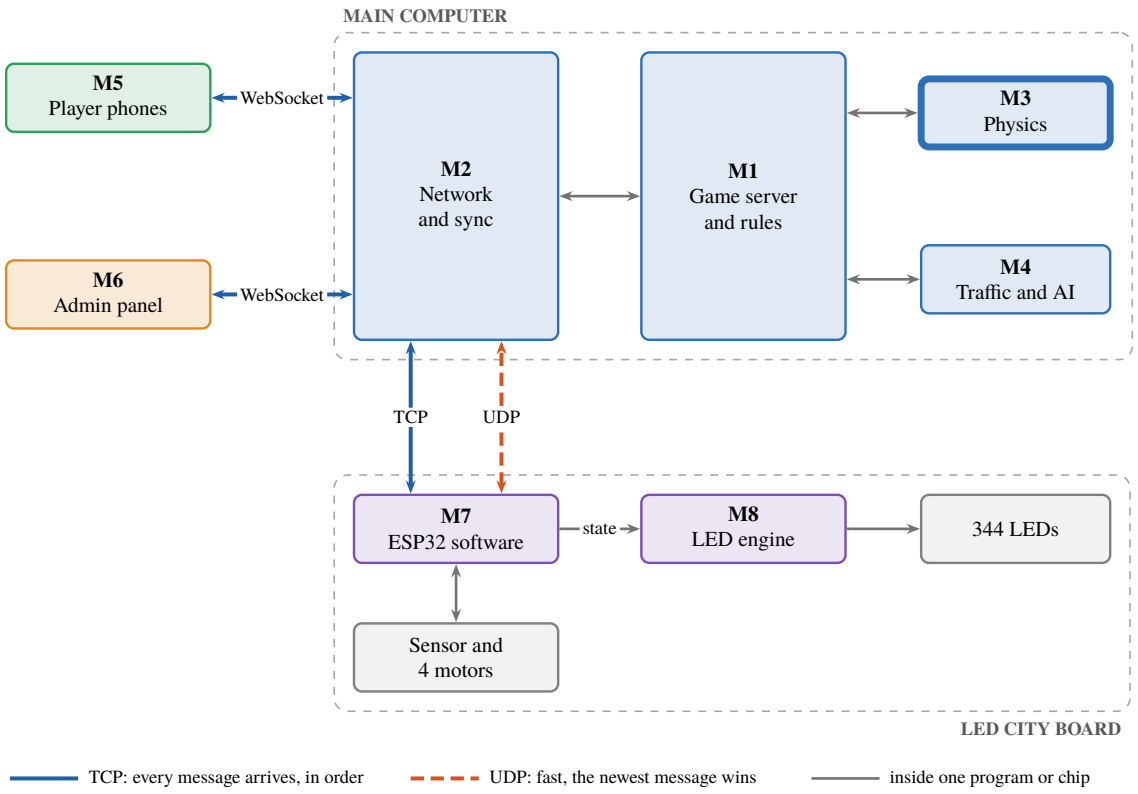

Figure 1: Where M3 sits in the system. M1 calls M3 in every tick; M5 uses a copy of M3 on the phone.

2. Requirements covered

ID	Requirement	What M3 does for it
R3	Driving and controls	Throttle, brake, left and right; no reverse; U-turn by steering in about a 3 m circle at low speed
R4	Collisions and falls	Collisions with buildings, barriers, NPC cars and motorcycles; a fast hit makes the motorcycle fall
R6	Obstacles	Road work and accidents are solid; an accident has a slow zone; a puddle makes the motorcycle slide

ID	Requirement	What M3 does for it
R7	Earthquake	Side push and less grip inside the earthquake radius that M1 sends
R11	Single authority	The code gives the same result for the same input; the server result always wins; the phone only predicts

3. Inputs and outputs

Dir.	Data	From / to	Content
In	City plan	Shared file	Roads, lanes, blocks and buildings as fixed colliders
In	Controls	M1	Every tick: throttle, brake, steer of humans and bots; a speed factor for each player
In	Obstacles, NPC cars	M1, cars from M4	Obstacle type, road, lane, position; boxes of the NPC cars
In	Earthquake	M1	Centre, strength, radius; the players M1 decided to drop
In	Parameters	M6, through M1	Physics group: top speed, grip, fall speeds, puddle and earthquake effects
Out	Poses	M1	Position, direction, speed, lean, sliding, fallen
Out	Events	M1 and M4	Collisions (with what, how fast) and falls
Out	Predicted pose	M5	The player's own motorcycle before the server answers

4. Main components

Component	What it does	Related modules
Motorcycle model	Speed, steering, turning, grip and sliding	—
Colliders	Buildings and the outer wall from the city plan; obstacles as boxes, slow zones and slippery areas	M1
Collision solver and fall judge	Solves contacts in a fixed order and decides falls from the hit speed	M1, M4
Earthquake effect	Push and less grip inside the radius	M1
Predictor	Prediction and correction on the phone	M5

5. How it works

One physics step. The server runs 30 ticks per second, and M3 runs 2 sub-steps in each tick, so one sub-step is about 16 ms. Small steps keep fast motorcycles from passing through thin walls. Each sub-step does five things in a fixed order, as the figure shows. A fixed order and seeded random numbers mean that the same inputs always give the same result on the server and on the phone.

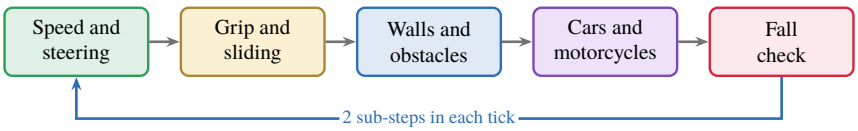

Figure 2: The five parts of one physics sub-step.

Driving. The throttle adds speed up to the top speed, and the brake and air drag take speed away. There is no reverse. The steering changes the direction. At low speed the motorcycle can turn in a circle of about 3 metres, so a U-turn is easy. At high speed the turn is wider. The model also gives a lean angle, which M5 uses to tilt the 3D motorcycle in a turn.

Grip and sliding. In a turn the motorcycle needs a side force. M3 compares the size of this force with the grip limit. If the force is below the limit, the motorcycle follows the turn. If it is above the limit, the motorcycle slides out of the turn. We compare only the size, so a left turn and a right turn act the same. On a water puddle the grip limit is very low, so the motorcycle slides but does not slow down. Inside an earthquake radius the grip is lower too, and there is a small push to the side.

Collisions and falls. Buildings, the outer wall and the road-work barriers come from the city plan and the obstacle list as boxes. NPC cars are boxes, and motorcycles are circles. To find collisions fast, M3 puts every object into a uniform grid, so it checks only objects in the same or next cell. A contact pushes the objects apart. If the hit speed is above the fall speed that the admin set, the motorcycle falls. When two motorcycles hit each other and their relative speed is above the limit, both of them fall. A fallen motorcycle is not solid, so it does not block the road for others. M1 gives the penalty time and later asks M3 to stand the motorcycle up again. An accident is solid in one lane and has a slow zone around it.

Prediction on the phone. On the phone, the player must see the own motorcycle move at once, not 100 ms later. So M5 runs a copy of M3 for this one motorcycle. Each control gets a number and is kept in a list. When a server state arrives, the phone takes the real position from the server, throws away the controls that the server has already used, and runs the remaining

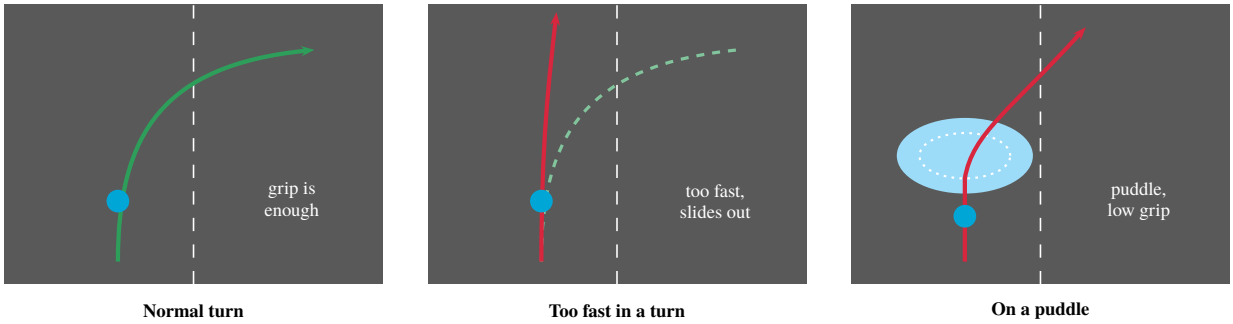

Figure 3: Three cases of grip, seen from above. Green is the wanted path; red is the real path.

controls again. If the new result is close to what the player sees, the difference is blended away in a short time. If it is far, for example after a crash, the motorcycle jumps to the right place.

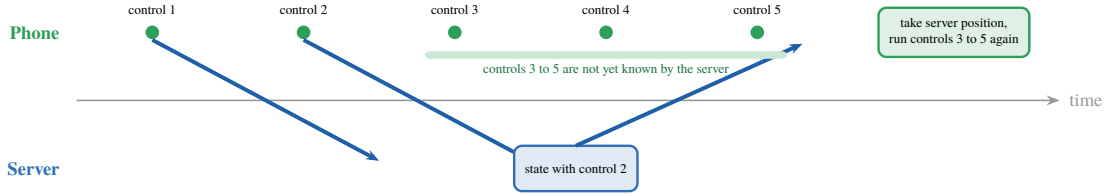

Figure 4: Prediction and correction for the player's own motorcycle.

6. Testing and demonstration

M3 needs no other module. A test makes a world from the city plan and gives scripted controls. We test a straight run and braking, a U-turn, slides to the left and to the right, wall and car crashes, the earthquake radius, the same result for the same inputs, and prediction with delay and loss. In the demonstration one motorcycle drives in the city with the keyboard. The admin changes grip and top speed, and we see the motorcycle change at once.